

Priyanga B

📍 Kerala, India | ✉️ priyangagbr@gmail.com | 🌐 priyangahere | 🎧 Priyanga-babu-rajendran | 🌐 Portfolio

SUMMARY

Data Science postgraduate researcher working at the intersection of machine learning, quantum computing, and trustworthy AI. Built ML pipelines, distributed analytics systems, and hybrid AI models. Current research spans privacy-preserving federated learning, quantum machine learning, and secure and reliable AI systems, with emphasis on privacy, robustness, and real-world deployment.

EDUCATION

M.Sc. Data Science — Vellore Institute of Technology, Chennai <i>CGPA: 8.34/10</i>	2025–2027
B.Sc. Physics — Sree Krishna College, Guruvayur, Kerala <i>CGPA: 7.967/10</i>	2021–2024
Higher Secondary — Biology Science with Maths — Kerala State Board <i>Percentage: 99%</i>	2019–2021

RESEARCH INTERESTS & TECHNICAL SKILLS

AI & Machine Learning: Deep Learning, NLP, Few-Shot Learning, Model Robustness, Statistical Modeling
Privacy-Preserving & Trustworthy AI: Federated Learning, Secure Distributed Learning
Security & Quantum ML: Network Intrusion Detection, AI Security, Quantum Neural Networks, Quantum Kernels, Hybrid Quantum-Classical Learning
Languages: Python, R, SQL
ML & Quantum Frameworks: PyTorch, Scikit-learn, OpenCV, PennyLane, Qiskit
Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn
Distributed Computing: Hadoop, HDFS, YARN, MapReduce, Spark
Tools: Git, GitHub, Jupyter, Streamlit, LaTeX, VS Code

PUBLICATIONS

Computational Methods for Multilingual Sentiment Analysis in Social Media: A Systematic Review (*Under review, IEEE Access*)

- Conducted a PRISMA-guided systematic review of multilingual sentiment analysis across classical ML, deep learning, and transformer-based approaches.
- Benchmarked BERT and XLM-R performance and methodological trends in multilingual and code-mixed settings.

RESEARCH PROJECTS

Adaptive Personalized Hybrid Quantum-Classical Federated Learning for Privacy-Preserving Intrusion Detection (*Ongoing*)

- Designing a personalized hybrid quantum-classical federated learning framework for network intrusion detection under heterogeneous, non-IID client data.
- Integrating differential privacy, quantum neural networks, and reliability-aware adaptive aggregation to improve privacy, robustness, and personalization.

Quantum Kernel-Based Few-Shot Auditing for Real-Time Contextual-Integrity Violation Detection in LLM Agent Outputs (*Ongoing*)

- Developing a quantum kernel-based few-shot anomaly detection framework to flag contextual-integrity violations in LLM agent outputs in real time.
- Designing quantum feature encodings and fidelity kernels with kernel SVM; benchmarking against classical kernels and LLM-as-judge baselines on accuracy, latency, label efficiency, and cross-benchmark generalization.

APPLIED PROJECTS

SignTalk – Continuous Word-Level Indian Sign Language Recognition | 🌐 Repo

- Built a real-time ISL recognition and gesture-to-text pipeline combining YOLOv5, MediaPipe landmarks with SVM, and NLP-based grammar correction.
- Achieved ~97% classification accuracy on a custom 82-word ISL dataset.

Distributed Web Log Analytics Using the Hadoop Ecosystem | 🌐 Repo

- Deployed a 3-node Hadoop cluster (HDFS, YARN, MapReduce, Spark) for distributed analysis of large-scale NASA web server logs.
- Implemented parallel analytics pipelines and a Streamlit dashboard for interactive visualization.

CERTIFICATIONS

- Introduction to Quantum Information | KAIST | Coursera
- IQM Quantum School Certificate | IQM Quantum Computers
- Data Science | IBM | Coursera
- Python for Everybody | University of Michigan | Coursera
- Cloud Computing | NPTEL
- Agentic AI and Autonomous Intelligent Systems Workshop | VIT, Chennai

LANGUAGES

English (Advanced) | Hindi (Advanced) | Tamil (Advanced) | Korean (Beginner) | Malayalam (Native)